

REAL TIME AGRICULTURAL COMMODITY DYNAMICS AND STABILITY TRACKER 21 April 2026

Objective:

This project involves cleaning, transforming, and analysing raw data using Excel and Power BI Process and creating an interactive Power BI dashboard to derive meaningful business insights.

The key goals of this project are:

1.Price Monitoring Track minimum, maximum, and modal prices across districts and markets to identify fluctuations.

2.Volatility & Stability Analysis Classify commodities into *Volatile* and *Stable* categories using the Price Stability Index.

3.Commodity Benchmarking Compare commodities (e.g., Wheat, Rice, Potato) across multiple districts and grades for performance insights.

4.Market Comparison Highlight differences in price ranges and stability between APMCs across states.

5.Grade & Variety Impact Assess how commodity grades (FAQ, Local, Medium, etc.) and varieties influence stability and price spreads.

6.Data- Driven Decision Support Provide actionable intelligence for farmers, traders, and policymakers to optimize supply chain analysis.

What is the project about?

The project takes **market arrival data** (commodity, variety, grade, state, district, market, arrival date, and prices) and transforms it into **visual insights**.

What problem are you solving?

This project is about **solving the problem of hidden insights in agricultural data**. By analyzing and visualizing it in Power BI, you're making the data **transparent, comparable, and useful** for real-world decision-making.

Why is this analysis important?

This analysis is important because it **bridges the gap between data and decision-making**. Instead of just listing prices, you're showing patterns, stability, and profitability making the information useful for farmers, traders and policy makers alike.

Data Sources:

Source Description and Timeline: Data government/Google dataset search 2013 - 2026

Domin: Agricultural economics/Market intelligence/Business Analytics

4. Attributes:

Attribute Name	Data Type	Description
State	Text/String	Name of the state where the market is located (e.g., Punjab, Uttar Pradesh).
District	Text/String	District within the state (e.g., Mohali, Mirzapur).
Market	Text/String	Specific APMC market where commodities are traded (e.g., Kharar APMC).
Commodity	Text/String	Name of the crop/product being sold (e.g., Onion, Tomato, Wheat).
Variety	Text/String	Sub-type or category of the commodity (e.g., Other, Local, Medium).
Grade	Text/String	Quality classification of the commodity (e.g., Non-FAQ, Local, Medium).
Arrival_Date	Date	Date when the commodity arrived in the market (format: DD/MM/YYYY).
Min_Price	Integer	Minimum price recorded for the commodity on that date (₹).
Max_Price	Integer	Maximum price recorded for the commodity on that date (₹).
Modal_Price	Integer	Most frequently quoted or average price for the commodity (₹).

5. Tools & Technologies:

EXCEL: Data extract, transform, load for cleaning and reshaping data before enter into data modelling and visualization.

Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (EXCEL)

Data Pre-processing – Cleaning, renaming columns, handling missing values, and creating derived features (Price Range, Stability Index).

Tasks Performed:

1. Data Collection & Import

- Import the dataset (Excel/CSV) into Power BI.
- Verify that all columns (State, District, Market, Commodity, Variety, Grade, Arrival_Date, Min_Price, Max_Price, Modal_Price) are loaded correctly.

2. Column Renaming & Standardization

- Ensure consistent naming conventions for clarity.
- Rename technical column names like Min_x0020_Price → **Min_Price**, Max_x0020_Price → **Max_Price**, Modal_x0020_Price → **Modal_Price**.

3. Data Type Conversion

- Convert **Arrival_Date** to Date format (DD/MM/YYYY).
- Convert **Min_Price**, **Max_Price**, **Modal_Price** to Integer/Number type.
- Keep **State**, **District**, **Market**, **Commodity**, **Variety**, **Grade** as Text/String.

4. Handling Missing Values

- Check for null or missing entries in any column.
- Fill missing categorical values with “Other” or drop rows if critical data is missing.
- For numeric fields, decide whether to impute (e.g., average) or exclude.

5. Removing Duplicates

- Identify duplicate rows (same State, District, Market, Commodity, Arrival_Date).
- Remove duplicates to avoid skewed analysis.

6. Data Consistency Check

- Ensure price values make sense (e.g., $\text{Min_Price} \leq \text{Modal_Price} \leq \text{Max_Price}$).
- Validate that Arrival_Date is in proper chronological order.

7.Data Modelling and DAX (Power BI)

Data Modelling – Designing a star schema with fact and dimension tables for efficient analysis.

● **Fact Table (Market_Fact):**

- Contains numeric measures like Min_Price, Max_Price, Modal_Price, Arrival_Date, Markert_ID, Commodity_ID, Price Range, Price Stability Index.

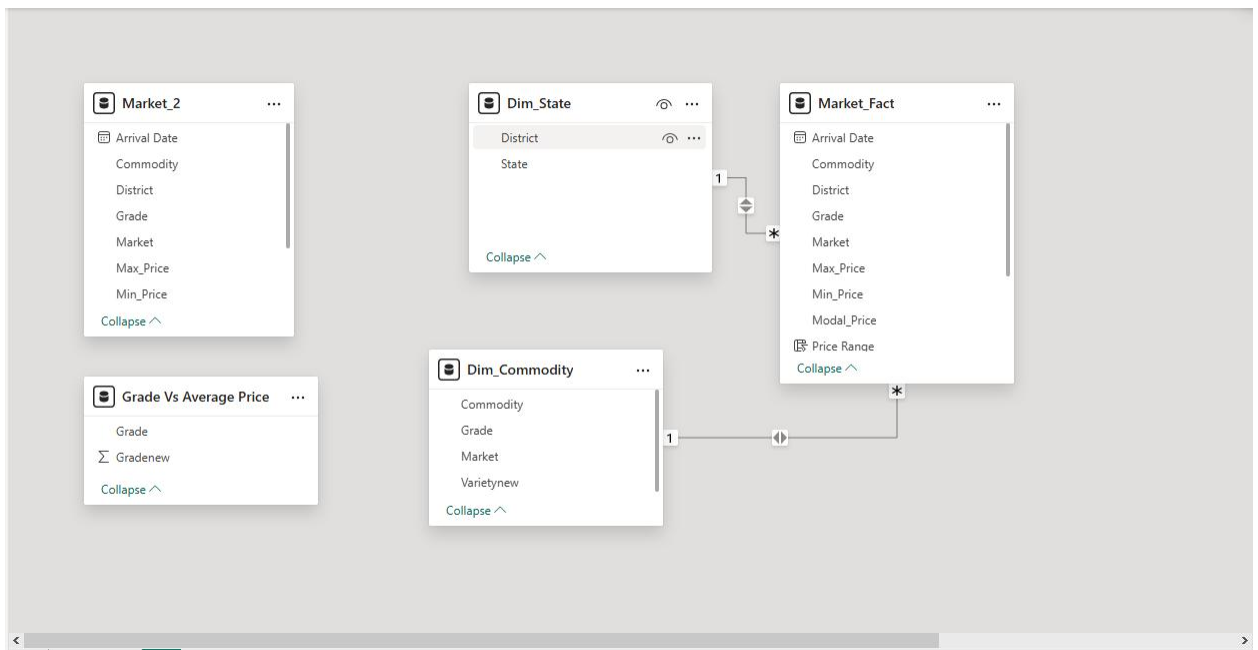
- Contains Categorical datas like State,Stability Category,Market Status.
- **Dimension Tables:**
 - **Dim_State** → State, District
 - **Dim_Commodity**→ Market, Commodity, Varietynew, Grade

Relationships

- **Dim_State** → **Market_Fact** (one-to-many)
- **Dim_Commodity** → **Market Fact** (one-to-many)

This star schema makes it easy to slice and filter data by state, district, market, or commodity,

SCREENSHOT OF DATA MODELLING



Calculated Columns & DAX Measures:

Create new calculated columns:

1. **Price Range** = $\text{Max_Price} - \text{Min_Price}$.

2. **Price Stability Index** = DIVIDE(('Market Price_Fact'[Max_Price]-'Market Price_Fact'[Min_Price]),'Market Price_Fact'[Modal_Price],0)- to compare stability across commodities.

DAX is used in Power BI to create calculated columns, measures, and custom KPIs.

Create New Column:

1. To create Stability Category”

DAX Measures:

Creating calculated measures (Price Range, Stability Index, Average Modal Price, KPI’s using CALCULATE).

1. Total Commodity
2. Average Price
3. Average Stability
2. Highest volatility
3. Market Entries
4. MAX Spread
5. Stable Item Using CALCULATE DAX Function
6. Volatile Item Using CALCULATE DAX Function
7. Volatility Rank by Commodity Using CALCULATE DAX Function

KPIs:

The KPI measures the relationship between the average price,volatility and stability of commodities across markets.

- .1. **Commodity Volume (126 Volatile / 193 Stable):** Monitors the total count of agricultural products, segmented by their price consistency to help identify market risk levels.
- 2.Volatility Avg Stability (0.16):** A specialized index that quantifies the overall fluctuation level across the monitored markets,where a lower number indicates higher general stability.
- 3.Market Entries (319):** Tracks the total number of data points or transaction records currently being processed in the real-time feed.
- 4.Market Entry Density:** A KPI tracking the number of active markets reporting daily versus the total available markets to assess the “completeness” of your real-time data.

8. Analysis and Visualizations (Power BI)

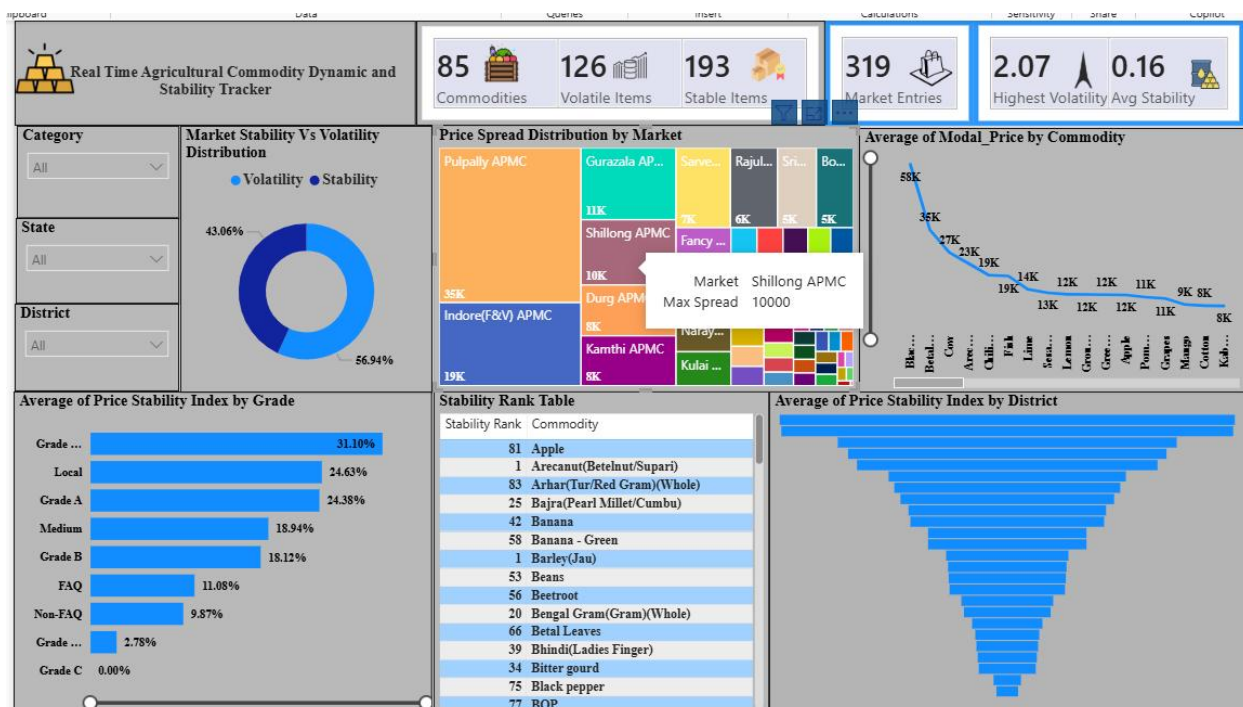
Dashboard Design – Building clustered column charts, line charts, and KPI cards to visualize insights.

Dashboard Features:

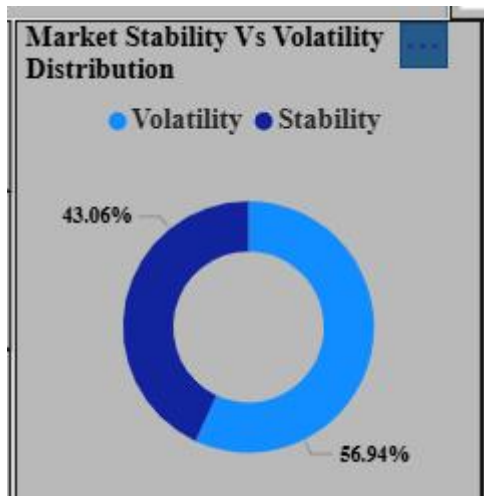
Visualization based on problem statement

1. **Tree Map Chart – Price Range by Market** Show Min vs Max prices for each commodity to highlight spread.
2. **Pie Chart – Stability vs Volatility Distribution** Percentage of commodities categorized as *Stable* vs *Volatile*.
3. **Funnel Chart– District vs Price Stability Index** Visualize which districts show higher volatility.
4. **Line Chart – Modal Price Trends by Commodity** Compare modal prices across commodities
5. **Clustered Column Chart – Grade vs Stability Category** Show how different grades (FAQ, Local, Medium, etc.) align with stability.
6. **Matrix Chart – Implementing a Ranking system** to evaluate the reliability of specific markets and commodities based on historical data

SCREENSHOT OF DASHBOARD



9. Insights based on the Market Stability Vs Volatility:



1. Descriptive Analysis (What happened?)

The chart shows **Volatility = 56.94%** and **Stability = 43.06%**. This means that in the observed market dataset, volatility dominates stability, with a **13.88% higher share**.

2. Diagnostic Analysis (Why did it happen?)

Agricultural or commodity cycles often drive volatility. Short shelf life and logistics issues amplify instability. Localized consumption patterns create uneven stability. Traders' behavior can push volatility beyond fundamentals.

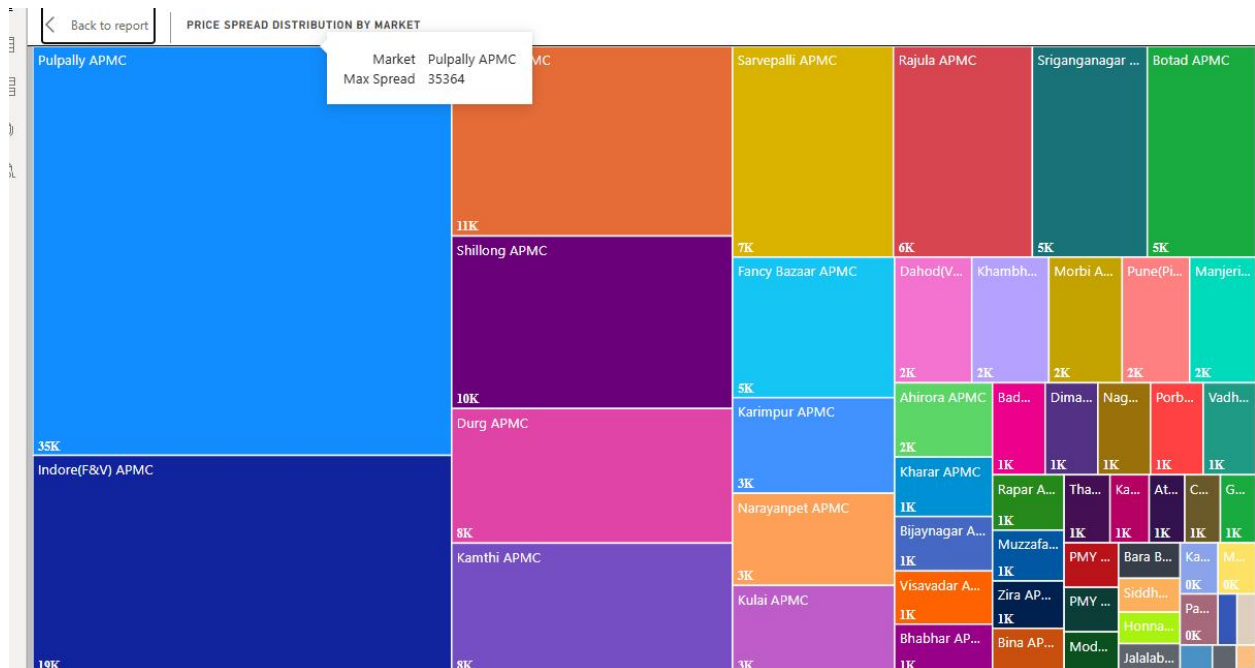
3. Predictive Analysis (What is likely to happen?)

If current conditions persist, **volatility will continue to outweigh stability** in the near term. Certain commodities may show **seasonal spikes** (e.g., harvest cycles, festival demand). Stability may improve in markets with **longer shelf life products** or **stronger demand predictability**.

4. Prescriptive Analysis (What should we do?)

Hedge against volatility using futures/options. Spread investments across both volatile and stable commodities. Build buffer stocks for volatile items to reduce supply shocks. Encourage cold storage, logistics improvements, and demand forecasting.

Insights based on the Price distribution across market:



1.Descriptive Analysis (What happened?)

The treemap shows wide variation in price spreads across markets. Pulpally APMC dominates with the highest spread (35K), followed by Indore (19K). Smaller markets like Fancy Bazaar sriganganagar contribute minimally to overall distribution.

2.Diagnostic Analysis (Why did it happen?)

High spreads in Pulpally and Indore suggest strong demand-supply imbalances. Regional factors, and market-specific demand drive volatility. Lower spreads in smaller APMCs may reflect limited trade volume or better price stability.

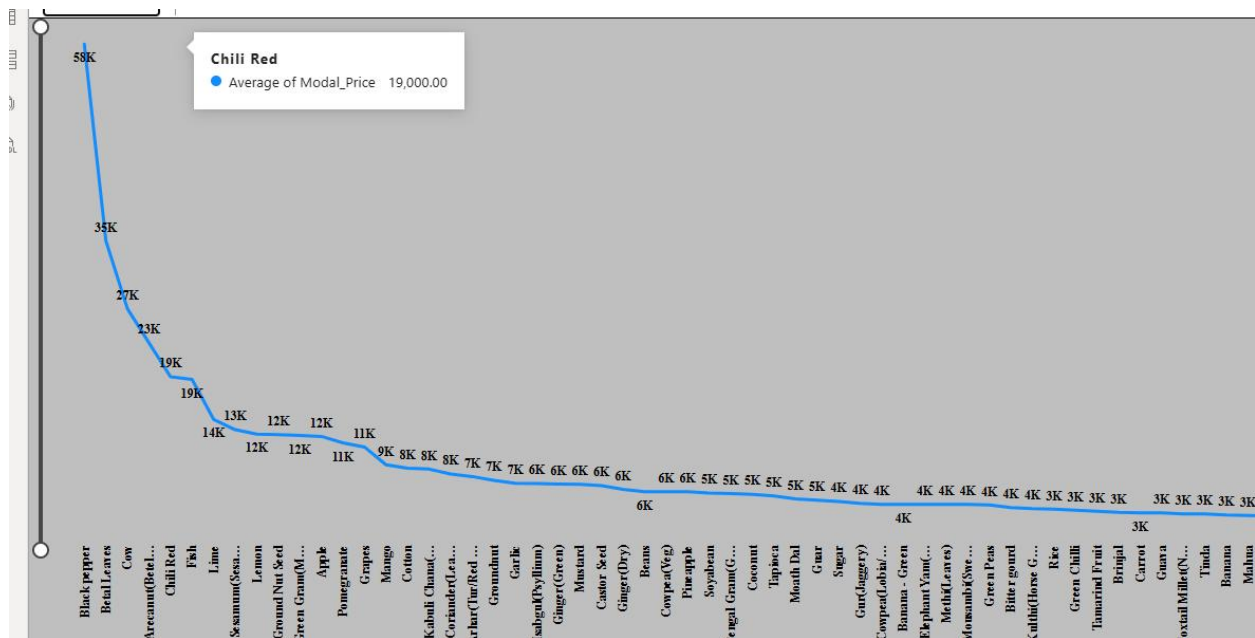
3.Predictive Analysis (What is likely to happen?)

Large markets will continue to show higher volatility due to scale and seasonal cycles. Smaller markets may remain relatively stable with modest fluctuations.

4.Prescriptive Analysis (What should we do?)

Focus monitoring and interventions on high-spread markets to reduce volatility. Encourage better logistics, storage, and demand forecasting in volatile regions. Leverage stable markets as benchmarks for improving efficiency across the network.

Insights based on the Modal price by commodity:



1. Descriptive Analysis (What happened?)

Banana shows the highest average modal price (58K), while commodities like Mango, Cotton, and Kela remain at the lowest (8K).

2. Diagnostic Analysis (Why did it happen?)

Price differences arise from factors such as seasonal demand, regional preferences, and commodity grade variations. High demand fruits like Banana command premium prices compared to staples like Cotton or Mango.

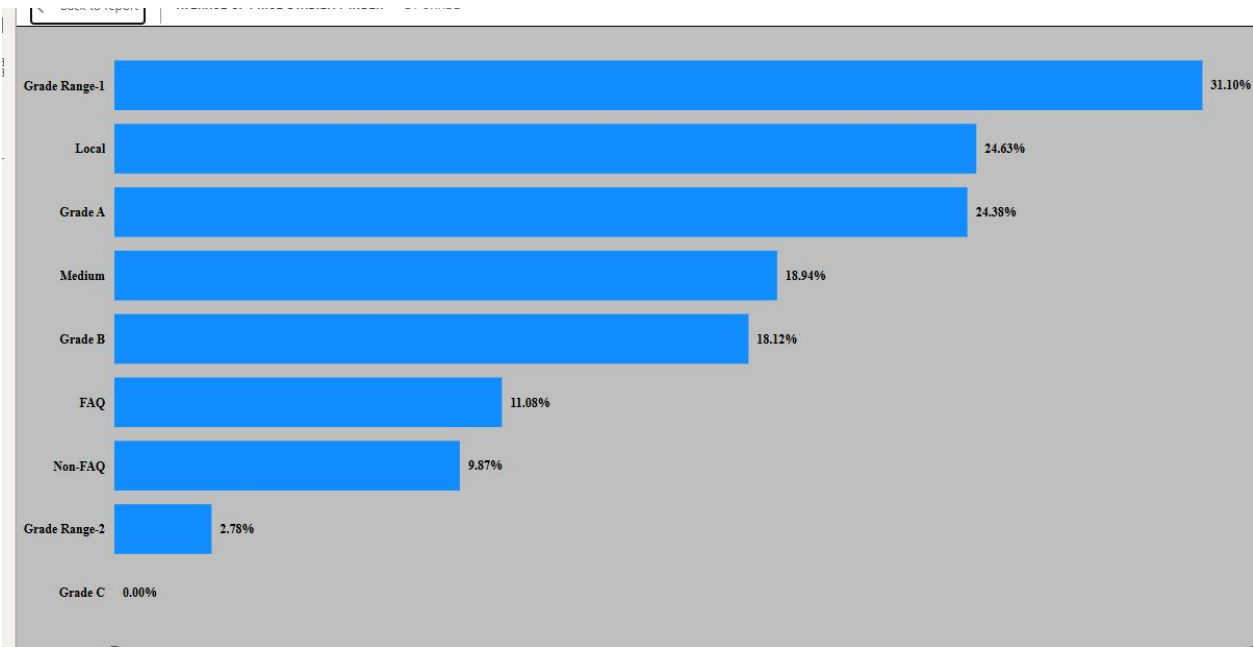
3. Predictive Analysis (What is likely to happen?)

Future trends suggest continued volatility, with premium fruits likely to maintain higher prices while staple commodities remain stable at lower ranges. Seasonal cycles may cause short-term fluctuations.

4. Prescriptive Analysis (What should we do?)

To optimize outcomes, traders should diversify across both high-value and stable commodities, monitor seasonal demand closely, and adjust procurement strategies to balance risk and profitability.

Insights based on the Grades:



1.Descriptive Analysis (What happened?)

Grade Range 1 shows the highest price stability (31.10%), while Grade C has no stability (0.00%).

2.Diagnostic Analysis (Why did it happen?)

Higher grades likely reflect consistent demand and quality, whereas lower grades suffer from variability and weaker market preference.

3.Predictive Analysis (What is likely to happen?)

Premium grades will continue to maintain stronger stability, while lower grades may remain volatile unless quality or demand improves.

4.Prescriptive Analysis (What should we do?)

Focus on strengthening demand for weaker grades through quality improvements, better marketing, and targeted distribution strategies.

Insights based on Stability Rank:

Stability Rank	Commodity
81	Apple
1	Arecanut(Betelnut/Supari)
83	Arhar(Tur/Red Gram)(Whole)
25	Bajra(Pearl Millet/Cumbu)
42	Banana
58	Banana - Green
1	Barley(Jau)
53	Beans
56	Beetroot
20	Bengal Gram(Gram)(Whole)
66	Betal Leaves
39	Bhindi(Ladies Finger)
34	Bitter gourd
75	Black pepper
77	BOP
52	Bottle gourd
45	Brinjal
73	Cabbage
1	Capsicum
71	Carrot
18	Castor Seed
60	Cauliflower
80	Chikoos(Sapota)
82	Chili Red
74	Cluster beans
26	Coconut
46	Coriander(Leaves)
31	Cotton
35	Cow
69	Cowpea(Lobia/Karamani)
62	Cowpea(Vag)

1.Descriptive Analysis (What happened?)

The table ranks commodities by stability, showing wide variation across items. Staples like barley and arecanut score very stable (rank 1), while apple and arhar are highly unstable. Overall, fruits and vegetables display mid-range volatility compared to grains and pulses.

2.Diagnostic Analysis (Why did it happen?)

High instability in apple and arhar reflects seasonal supply shocks. Stable ranks for barley and arecanut stem from consistent demand and longer shelf life. Mid-range commodities like banana and bhindi are influenced by regional demand and storage challenges.

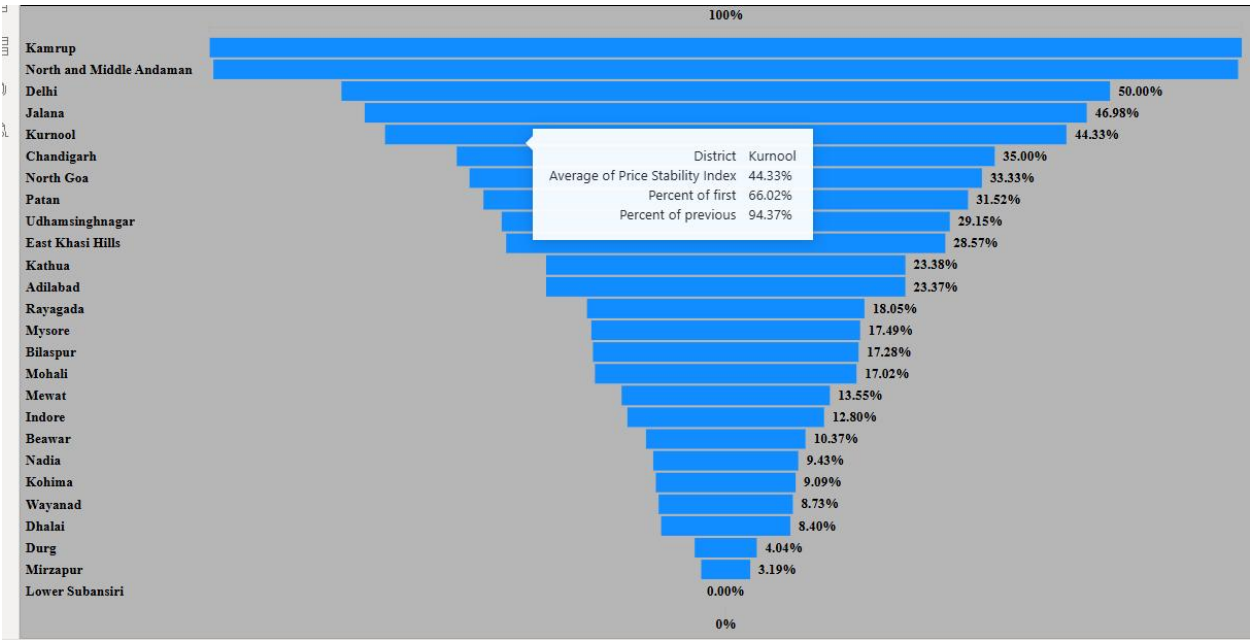
3.Predictive Analysis (What is likely to happen?)

Stable commodities will continue to show resilience in pricing and availability. Volatile items like apple and arhar are expected to fluctuate with seasonal cycles and weather. Vegetables in the mid-range may show moderate stability improvements with better logistics.

4.Prescriptive Analysis (What should we do?)

Focus on storage and supply-chain improvements for perishable, unstable commodities. Promote diversification by balancing stable grains with volatile fruits and pulses. Use predictive insights to plan procurement and pricing strategies ahead of seasonal demand shifts.

Insights based on the price stability by district wise analysis:



1.Descriptive Analysis (What happened?)

Districts show varying Price Stability Index averages, with some consistently higher and others lower, highlighting uneven market stability.

2.Diagnostic Analysis (Why did it happen?)

Differences arise due to seasonal demand, commodity regional consumption patterns, and grading standards impacting stability.

3. Predictive Analysis (What is likely to happen?)

Volatility is expected to persist in sensitive districts; targeted interventions like improved storage, balanced supply chains, and district-specific policy support can enhance stability.

4. Prescriptive Analysis (What should we do?)

Districts at the top of the chart demonstrate strong price stability. Their practices—such as efficient distribution networks, cooperative farming models, or effective market regulation—should be studied and adapted in weaker districts. Government and market regulators should design district-specific interventions—such as minimum support prices (MSP), targeted procurement, or localized trade policies—to stabilize vulnerable regions.

Overall key Insights & Conclusion:

Monitor commodity prices (Min, Max, Modal) across districts and markets. Identify volatility vs stability using the Price Stability Index. Compare grades and varieties to understand their impact on price behavior.

Price Range: Quantify spreads (e.g., Rice ₹3400–₹8100, Potato ₹400–₹600). **Stability Index:** Track % volatility (e.g., Rice 67.14%, Potato 40%). **Commodity Count:** Document stable vs volatile commodities (e.g., 70% stable, 30% volatile).

Use bar charts, pie charts, and scatter plots to visualize price and stability. Implement dashboards in Excel/Power BI for automated monitoring. Apply filters by district, commodity, and grade for actionable insights.

Supports farmers, traders, and policymakers in decision-making. Helps stabilize volatile commodities (e.g., Rice, Potato, BOP). Provides benchmarks for regional trade and supply chain optimization.

Daily updates to capture short-term volatility. Seasonal tracking for commodities with cyclical demand (e.g., Wheat, Chilli).

10. Final Conclusion:

This project proves that **data-driven decision support is essential for agricultural markets**. With real-time monitoring, stakeholders can **reduce risk, stabilize prices, and improve profitability**. Ultimately, the dashboard is not just a visualization tool—it is a **strategic framework for sustainable and transparent agricultural market management**.

